

Nour Belgacem

Embedded Systems Engineer

C / C++ / Qt / Linux

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Tunis, Tunisia

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EXPERIENCE

- **Development of a Multiprotocol Interface for a Test Bench in Qt/C++** *February 2026 – July 2026*

  Graduation Project Internship SERMA International, Tunisia

  - Development of graphical interfaces in Qt/C++ for memory data acquisition (RAM/FRAM) via the RS-232 protocol and real-time communication over the CAN/CAN FD bus.
  - Design of a dual-process software architecture with IPC communication to eliminate latency and ensure sub-millisecond timing accuracy.
  - Development of a qualification interface for an aeronautical equipment integrating signal acquisition via the NI USB-6001 board and automatic test report generation.
  - Project management using the Scrumban methodology with task tracking on Jira within a multidisciplinary team.

  Key skills : Qt/C++, CAN/CAN FD, RS-232, IPC, NI USB-6001, Embedded Systems, Aeronautics, Scrumban, Jira.

- **Design of an Electrical Cable Break Detection Board** *June 2025 – July 2025*

  Engineering Internship Electrical Systems Laboratory, ENIT

  - Complete design of an electronic board for cable break detection, from the study of the electronic schematic to PCB routing and physical prototype fabrication.
  - Development of the embedded firmware of the PIC12F675 microcontroller in C, using MPLAB IDE, and code transfer to the component via the PICkit programmer.

  Key skills :PIC12F675, C Language, MPLAB IDE, PICkit, Oscilloscope.

ACADEMIC PROJECTS

- **Implementation and Fabrication of an Autonomous Fire Fighting Robot with RTOS** *September 2025 – January 2026*

  Academic Project ENIT, Tunisia

  - Complete design of an autonomous fire fighting robot: definition of the specifications, component selection, prototype fabrication integrating flame sensors, drive motors and water pump.
  - Development of the embedded firmware on STM32F407 under STM32CubeIDE with FreeRTOS, structuring autonomous movement, flame detection and pump control as independent multitasking tasks.

- **DC Motor Speed Control with BeagleBone** *January 2025 – May 2025*

  Academic Project ENIT, Tunisia

  - Development of an application in embedded C under embedded Linux on BeagleBone for DC motor speed control through generation of MLI/PWM signals from a potentiometer input.

EDUCATION

- **National Engineering School of Tunis (ENIT)** *September 2023 – June 2026*

  Engineering Degree in Electrical Engineering – *Advanced Electronic Systems (SEA)* Tunis, Tunisia

TECHNICAL SKILLS

- **Programming:** C, C++
- **Software and Embedded Development:** Qt Creator, Qt Framework, STM32CubeIDE, Code::Blocks
- **Embedded Systems:** STM32, PIC, BeagleBone, FreeRTOS, Embedded Linux
- **Simulation Tools:** LTspice, PSIM, Proteus, ModelSim, MATLAB/Simulink

LANGUAGES

French: B2 (DELF) **English:** B2 (TOEIC: 785)